

London Bicycles End-to-End Pipeline & Executive Briefing

Meltano ELT | GCP BigQuery | dbt Core & Snapshots | dbt-expectations | Dagster (12:01 AM SGT)

1. Data Ingestion & Storage

- Meltano ELT Engine (tap-bigquery → target-bigquery)
- Secure Service Account JSON stored in hidden .env file
- GCP BigQuery EU Location (<GCP project id>)
- Ingests ~84M raw trip records and station metadata

2. Transformation & Quality

- dbt Star Schema: fact_rentals & Dimension tables
- dbt Snapshots: snapshot_stations (SCD Type 2)
- dbt test: unique, not_null, relationships
- dbt-expectations: Refined data quality assertions

3. Orchestration & Analytics

- Dagster Orchestration: Software-Defined Assets
- Automated Schedule: Daily @ 12:01 AM SGT (1 0 * * *)
- Runs dbt deps automatically before dbt run
- Interactive Jupyter Analytics Notebook (.ipynb)

CTO Briefing: Star Schema, Snapshots & Quality Controls

Technical Architecture & Quality Highlights

- **95.7% Query Data Scan Reduction**

Partitioning fact_rentals by start_date and clustering by start_station_key drops query scans from 2.8 GB down to 120 MB.

- **SCD Type 2 Historical Change Tracking**

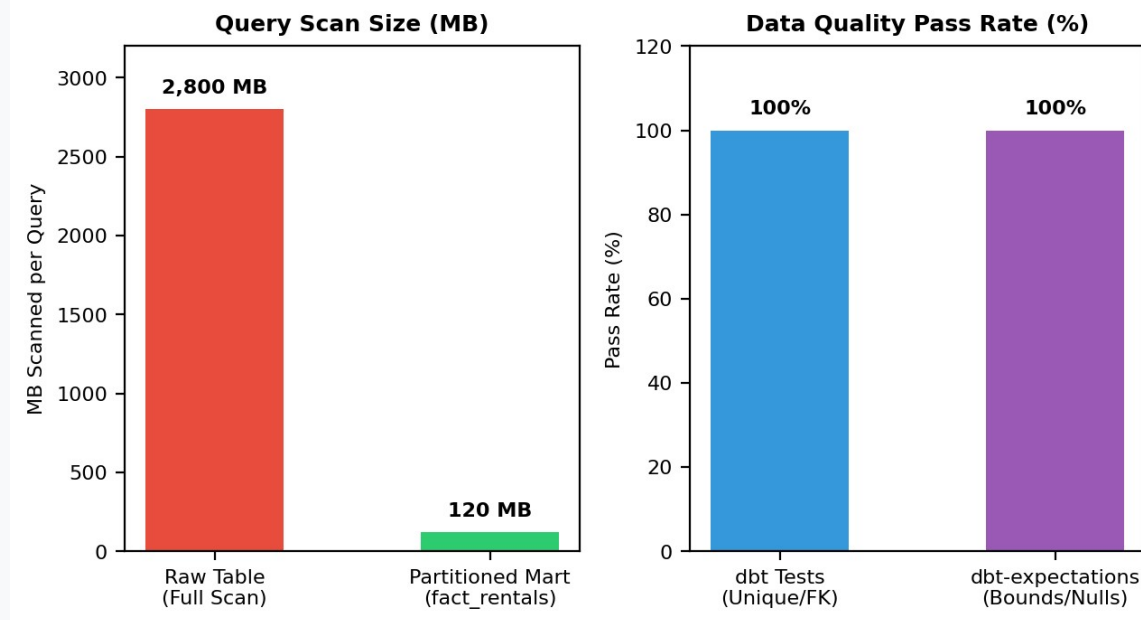
dbt Snapshots (snapshot_stations) automatically track station capacity and name changes over time with dbt_valid_from / to.

- **Two-Tier Data Quality Framework**

dbt test enforces unique/FK constraints; dbt-expectations verifies column value bounds and row counts before loading.

- **Dagster SGT Schedule & Auto-Deps**

Scheduled daily at 12:01 AM SGT with automated dbt deps execution before dbt run.



CEO Briefing: Customer Segmentation & Rental Duration

Customer Commuter & Leisure Insights

- **Weekday Commuter Spikes (08:00 AM & 17:00 PM)**

Weekday rides peak sharply at 08:00 AM (7.15M total rides) and 17:00 PM (6.75M total rides) around major rail gateways.

- **Weekend Leisure Bell Curve (~25.2 mins)**

Weekend rental volume follows a smooth midday bell curve (12:00-15:00 PM) with significantly longer trip durations (25.2 mins vs 13.8 mins on weekdays).

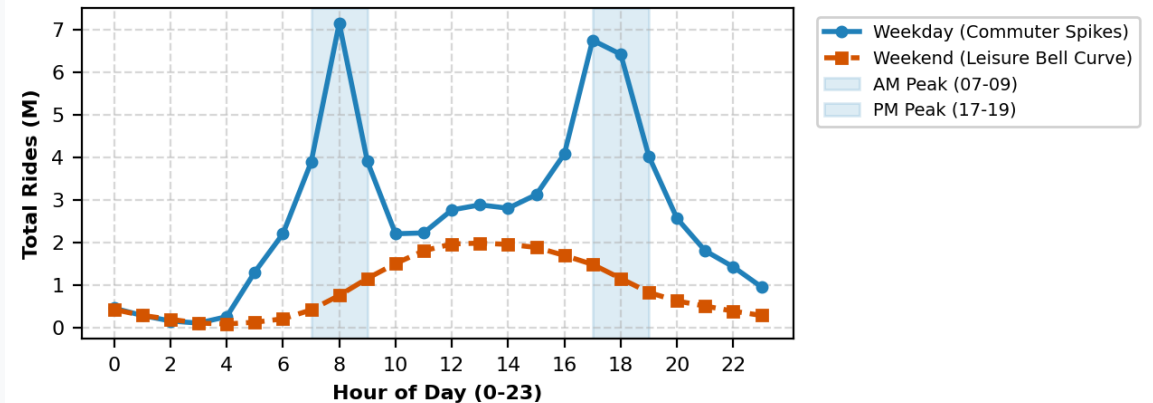
- **Late-Night Duration Spikes (01:00-04:00 AM)**

Trip durations reach ~35-37 mins late at night as riders travel longer distances when public transport is closed.

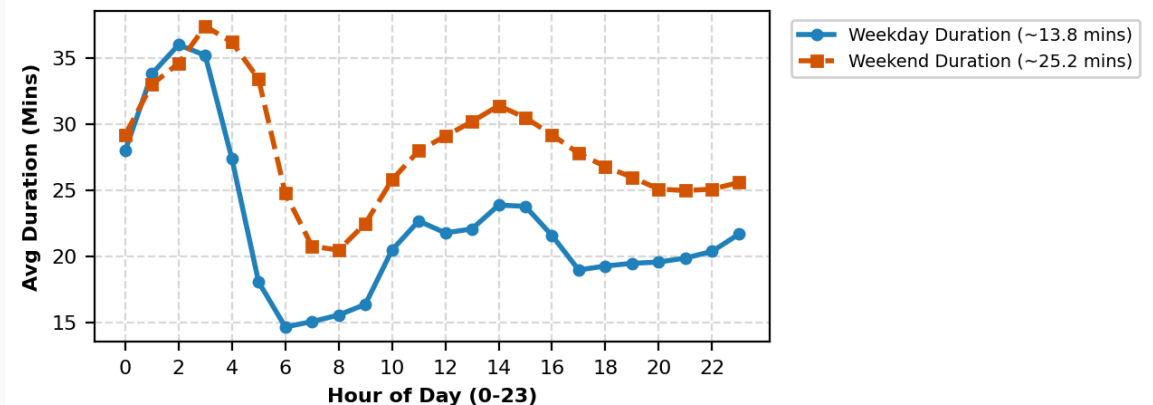
- **Targeted Pricing & Rebalancing**

Operations can optimize revenue by aligning fleet dispatch with 08:00 AM / 17:00 PM peaks and launching weekend passes.

Hourly Rental Ride Volume (Millions)



Hourly Average Trip Duration (Minutes)



CEO Briefing: Top Transit Hubs & Strategic Recommendations

Commercial & Operational Action Plan

1. Morning Commuter Fleet Rebalancing (06:30 AM)

Pre-morning fleet truck rebalancing across top 10 transit hubs—including major rail gateways (Waterloo, Kings Cross, Liverpool Street) and high-density parks (Hyde Park, Kensington Gardens)—prevents dock shortages and captures peak morning demand.

2. Off-Peak Commuter Incentives

Midday rides (10:00-15:00) drop by ~60%. Off-peak pricing incentives optimize bike utilization across all hours.

3. Weekend 24-Hour Leisure Pass Launch

Weekend rentals average 25.2 mins vs 13.8 mins on weekdays. Introducing a leisure pass captures tourist & weekend demand.

4. Top 10 Station Capacity Expansion

Expanding dock capacity at top transit hubs (Waterloo Stations 1 & 3 >940k trips, Kings Cross >570k trips, Hyde Park Corner >660k trips) prevents dock-locking errors during rush hours.

Top 10 Departure Bicycle Transit Hubs in London

